# Advanced prompting vs LoRA fine-tuning for safety-critical LLM applications: A comparative study on cosmetology product recommendations

NLP

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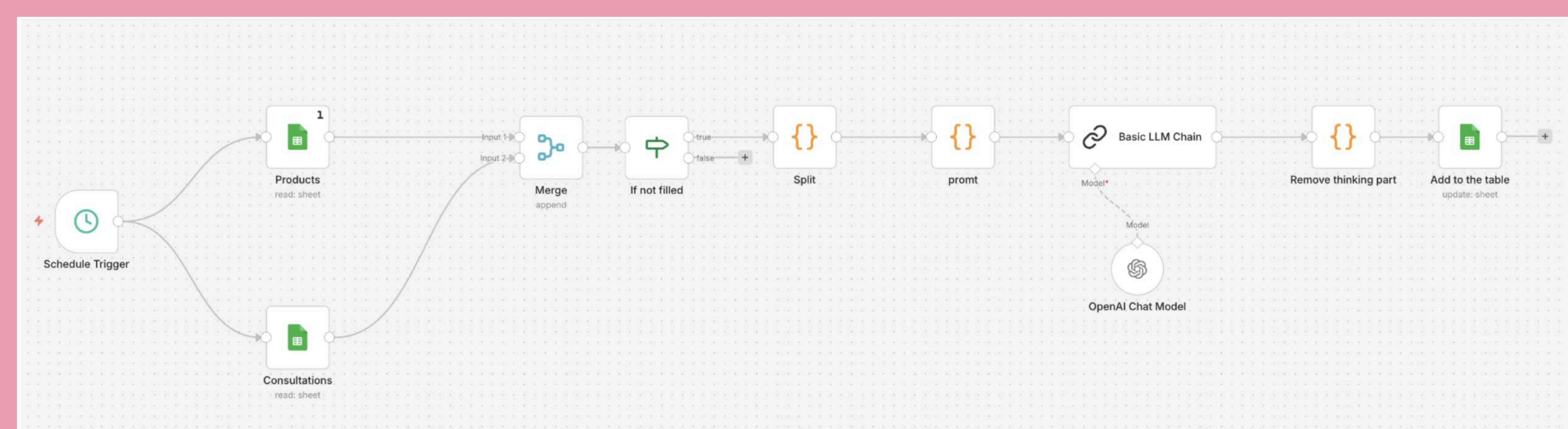
## Motivation

Safety-critical LLM applications (medical, cosmetic, financial recommendations) require strict adherence to constraints: budget limits, allergy restrictions, product availability. This study investigates whether advanced prompting can match fine-tuning quality on personalized cosmetology recommendations while avoiding hallucinations and constraint violations.

## Dataset




## Setup

**Model**

Mistral-7B-Instruct

**Training**

55 expert consultations (80/20 split)

**LoRA**

r=16, 5 epochs, gradient checkpointing

**Baselines**

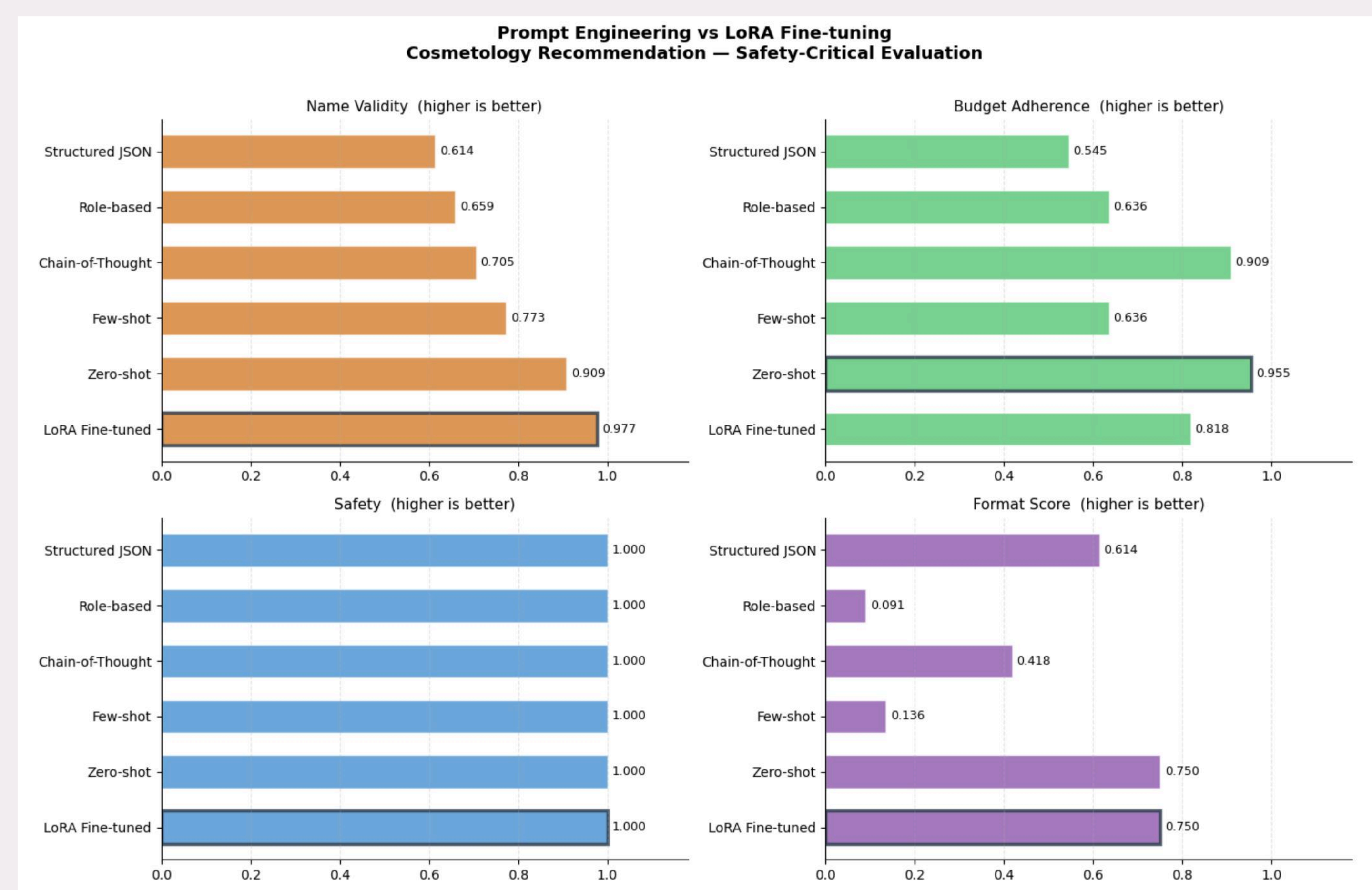
5 prompt variants (zero-shot, few-shot, CoT, role-based, JSON)

**Metrics**

Name Validity, Budget Adherence, Safety (allergens), Format Consistency, Inference Time

**Test**

11 diverse skin profiles with ground truth recommendations




| Method | Name Validity | Budget | Safety | Format Score | Inference (s) |
|---|---|---|---|---|---|
| LoRA Fine-tuned | 0.977 | 0.818 | 1.000 | 0.750 | 75.2 |
| Zero-shot | 0.909 | 0.955 | 1.000 | 0.750 | 75.3 |
| CoT | 0.705 | 0.909 | 1.000 | 0.418 | 73.6 |
| Few-shot | 0.773 | 0.636 | 1.000 | 0.136 | 77.2 |
| Structured JSON | 0.614 | 0.545 | 1.000 | 0.614 | 33.4 |
| Role-based | 0.659 | 0.636 | 1.000 | 0.091 | 75.2 |

**Role-based prompting destroys output format**

Prompt-heavy instructions collide with the model's safety filters, producing refusal artefacts like repeated tokens, clinical-style monologues, and structurally corrupted text.
products are frequently omitted altogether, and budget parsing fails in the vast majority of cases.
finding: instruction-only guardrails are unreliable; robust safety requires explicit, rule-based output validation.

## Results

**LoRa fine-tuning leads product-name accuracy**

LoRA achieves near-perfect identification of real-world cosmetic products, with no allergen violations.
Budget adherence is solid but falls slightly behind the best prompting method.
Training cost remains negligible, under 17 minutes of GPU time.

**Zero-shot wins budget control and matches lora on format**

Zero-shot delivers the strongest budget obedience across all methods, while its format quality now equals lora's.
product names are correctly recognised in the vast majority of cases, and safety remains flawless — all with zero training overhead.

**Few-shot prompting corrupts output structure**

Template-text leakage, emoji injection, and partial repetition of few-shot examples break the expected structure.
budget figures become unparseable in most responses, rendering the output practically unusable.
finding: few-shot examples severely degrade performance on structured tasks and should be avoided.

**Structured json sacrifices quality for speed**

Strict JSON schemas produce the fastest inference but sharply reduce name validity, budget compliance, and overall output coherence.
The model frequently fills fields with generic phrases or omits them, increasing hallucination.
Finding: Free-text generation followed by parsing outperforms raw JSON constraints in both accuracy and reliability.

**Chain-of-thought improves budget reasoning but hallucinates product names**

Reasoning through costs yields strong budget adherence, yet the process causes the model to invent or vaguely describe products.
name validity drops significantly; the method is only useful when paired with a post-hoc brand verification step.

## Conclusion

Use zero-shot as the production baseline for safety and budget constrained recommendations. Invest in LoRA only when exact brand recognition is essential; discard few-shot, role-based, chain-of-thought, and JSON-constrained approaches as unreliable.

## Key finding

Zero-shot prompting equals or surpasses LoRA on safety, format consistency, and budget adherence while requiring no training: complex prompting strategies consistently degrade output quality. The sole measured advantage of fine-tuning is higher product-name accuracy.